

Crisphine M. Ngari

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[Github Repository](#)

Profile Statement

Master's student in electrical engineering with strengths in analog and digital circuit design, RF systems and embedded hardware–software integration. Demonstrated research and teaching assistant experience in semiconductor device fundamentals and troubleshooting complex systems. Skilled in problem solving, documentation and collaboration, with a conference paper under IEEE CONNECT. Eager to apply AI-driven automation techniques to layout generation and optimization.

Education

Southern Illinois University- Carbondale

Master of Science, Electrical and Computer Engineering (GPA: 3.89/4.0)

Aug 2025 - Present

Illinois, USA

Gandhi Institute of Technology and Management

Bachelor of Technology, Electronic and Communication Engineering (GPA: 8.24/10)

Sep 2020 - May 2024

Bangalore, India

- **Achievements:** Recipient: Study in India Scholarship (Merit-Based) | 2020 – 2024

PC Kinyanjui Technical Training Institute

Diploma, Electrical and Electronics Engineering (Telecommunication)

Sep 2016 - Aug 2019

Nairobi, Kenya

Kivaa High School

Kenya Certificate of Secondary Education

Feb 2011 - Nov 2014

Machakos, Kenya

Research Experience

Indoor Wireless Localization Using Fingerprinting

Jul 2023 - Apr 2024

- Built a Wi-Fi fingerprint database using Python scripting and AI Tools, mapping RSSI measurements from multiple access points to known physical locations
- Applied the K-means algorithm in a Cadence SKILL–style workflow to match live RSSI readings to the database and performed root-cause analysis on signal anomalies

Real-Time Energy Monitoring Using Simulation Modelling

- Developed a discrete event simulation in Simulink and MATLAB scripts to model manufacturing energy consumption and optimize sensor sampling rates
- A module equipped with various sensors to collect the parameters to monitor and report back to the cloud-based database. A dashboard to capture key performance indicators for the executives.
- Implemented an AI-driven web diagnostic tool for frontline workers, integrating real-time data analytics to perform automated ‘first-aid’ on malfunctioning nodes

Distributed Health Monitoring System for Street Light Tracking

- Led the development of a distributed health monitoring system for tracking streetlights during the 2023 Smart India Hackathon.
- The realized system has functionalities to track and report KPIs of streetlight nodes in real-time to a central dashboard, remote debugging of faulty nodes and firmware updates via FOTA.

Federated self-supervised contrastive learning for unknown attack detection in IoT Networks

- Designing an algorithm to detect unknown attacks with fewer labeled samples while keeping raw traffic data local to each organization.

Professional Experience

Southern Illinois University-Carbondale | *Graduate teaching Assistant*

Aug 2025 - Present

- Grading assignments
- Assisting students with homework
- Setting lab practical.
- Conducted research on semiconductor device physics, including MOSFET and BJT operation, assisting in lab experiments using root-cause analysis methods
- Other teaching duties as assigned

Afrotel Group Telecommunication Engineering | *Network Engineer*

Jun 2024 - Jul 2025

- Troubleshooting network issues using root-cause analysis to prevent downtime and ensure peak performance levels
- Installing and configuring hardware and software, including those for backups and system performance monitoring, to ensure optimal network performance.
- Modifying and upgrading network equipment to meet changing business requirements and user needs.
- Conduct site surveys to assess network coverage and identify areas where network performance needs optimization and enhancement.

- Installation and maintenance of a Fiber Optics FTTX infrastructure
- LTE,5G, Radio Access Networks (RAN), Inside Building Solutions (IBS), and Microwave installation and maintenance.
- Connects a mobile network's core network, service network, radio access networks, and external networks to the Mobile Packet Backbone Network (MPBN)
- Managing Multiprotocol Label Switching (MPLS) data centers
- Perform regular FTTX tests, OTDR, CD/PMD, ORL, Power Meter, and scopes.
- Perform fiber optic maintenance including Point of break /link analysis, splicing, core alignment and splice quality validation using necessary tools and techniques
- Evaluates, tests, and repairs old or malfunctioning fiber network infrastructure along with scheduled preventive maintenance routines covering: Optical cable, Closure kits, Ducts and Manholes
- Ensure Fiber networks (OSP, ISP, OHL) are planned, designed, deployed and As-builts done to agreed standards.

EduSkills Foundation | Intern

May 2023 - Jul 2023

- Identified, troubleshot, analyzed, and solved technical challenges related to embedded systems and software, performing root-cause analysis on circuit issues involving resistors, capacitors, and diodes
- Integrating software with embedded hardware components, such as sensors, actuators, and communication modules, using programming skills to optimize performance
- Participating in the design, development, and testing of embedded software and hardware solutions, leveraging AI tools to improve testing workflows

South Eastern Kenya University | ICT Technician

Sep 2019 - Feb 2021

- Maintained CCTV systems for security.
- Installed, upgraded and managed network infrastructure including software and hardware.
- User (students and staff) training of ICT systems, EPR system support, installation and connectivity
- Managed computer labs and conducted ICT training for students and staff.
- Network monitoring and administration, performing root-cause analysis to resolve connectivity issues using monitoring tools

Jomo Kenyatta International Airport | Intern

Sep 2017 - Dec 2017

- Installed crash alarm systems and maintained PA-BX and associated equipment.
- Replaced baggage handling motors and provided installation, repair, and service for telephone equipment.
- Maintained and repaired flood lighting systems, electrical installations
- Performed electronics repair and maintenance for radar systems and radio signaling equipment.

Skills

- **Circuit Design & Semiconductor Physics:** Analog Circuit Design, Digital Circuit Design, PCB circuit Designing, Semiconductor Device Physics, MOSFET, BJT, Resistor, Capacitor, Diode
- **Hardware & Embedded Systems:** Embedded systems design, Embedded systems hardware security, Arduino, Raspberry pi, NodeMCU
- **Programming, Tools & AI:** VHDL, Verilog, CPLDs, FPGAs, Cadence Virtuoso, Cadence SKILL, AI Tools, AI Agents- , Programming
- **Networking & Administration:** Network monitoring, administration
- **Soft Skills & Certifications:** collaboration, team playing, Leadership, Great communication, Time management, Planning, Organizing, Problem Solving, troubleshooting, Root-cause Analysis, analytical, interpersonal skills, Research, documentation, work at height (certified), confined spaces, all weathers

Organizations and Leadership

GITAM Directorate of External Relations | Students Coordinator

May 2023 - Feb 2024

GITAM INTERNATIONAL STUDENTS ASSOCIATION | President

Mar 2022 - May 2023

GITAM SPORTS DEPARTMENT | Students sports Associate

May 2021 - Jun 2022

PC Kinyanjui Technical Students council | Assist Minister of Finance

Jul 2018 - Aug 2019

Scholarly Achievements

- Study In India (SII) Merit based scholarship
- Conference paper publication under IEEE CONNECT

Languages

- English (Bilingual proficiency)
- Swahili (Native speaker)